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Ravi Pandey

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Education

University of Nebraska Omaha, USA **Jan 2025 — Dec 2026 (Expected)**

- *M.S. Computer Science (**GPA: 4.0**)*
- Courseworks: Design and Analysis of Algorithms, Artificial Intelligence, Database Management System, Advanced Operating System, Software Specifications & Design, Cloud Computing

Institute of Engineering, Pulchowk Campus, Tribhuvan University, Nepal **2018 — 2023**

- *B.E. Computer Engineering (**GPA: 3.72**)*
- Courseworks: Probability & Statistics, Calculus, Artificial Intelligence, Software Specifications, Data Mining, Big Data Technologies, Data Structures, Object Oriented Programming, Computer Networks, Theory of Computation, Computer Architecture

Experience

CIST Graduate Research Assistant **Jan 2025 — Present**

University of Nebraska Omaha, Omaha, USA

- **Designed and implemented extensions to a declarative programming language** by introducing functional syntax and custom operators, using **parser construction** and AST transformations to enable new expressive capabilities in Answer Set Programming.
- **Built a Python-based compilation pipeline using the Clingo API** to parse custom rules, extend solver AST nodes, and transform functional programs into standard ASP and back, enabling compatibility with existing solvers while preserving higher-level abstractions.

Associate Software Engineer **Jun 2023 — Aug 2024**

GrowByData Services, Lalitpur, Nepal

- **Built and integrated scalable data ingestion and tracking services** by developing API-driven integrations with Google Analytics, Google Ads, and Meta Ads, improving observability of user behavior and supporting data-driven decision-making for e-commerce clients.
- **Developed end-to-end ETL and analytics workflows** by extracting data from multiple backend services, cleaning and transforming high-volume datasets in Python, and delivering integrated dashboards and business intelligence insights to support customer-facing decision-making.

Game Development Intern **Feb 2023 — Jun 2023**

Avana Game Studios, Dallas, TX, USA (Remote from Nepal)

- **Developed interactive game features in Unity using C#**, created 3D assets in Blender and implemented a guitar simulation to map finger strumming to MIDI notes with audio validation against real instrument samples.

Skills

Programming Languages: Python, C, C++, Java, SQL, JavaScript
Frameworks & Tools: Numpy, Pandas, Matplotlib, Scikit-learn, RESTful APIs, Git, Power BI, Snowflake, Linux
Machine Learning/Deep Learning: NLP, Computer Vision, LLM, Statistical Analysis, ETL, Data Visualization
Web Development: Django, Rest APIs, Flask, Fastapi, Streamlit, HTML, CSS, Bootstrap, React
Database & Deployment: MySQL, MongoDB, Docker, AWS, Azure, CI/CD
Soft Skills: Problem solving, Good communication, Teamwork, Leadership, Time management
Communication: English, Nepali, Hindi

Publications

- D.C. N. & **Pandey, R.**, et al. (2023). *Improved Pre-processing Techniques for Mobile-based OCR Systems*. In Proceedings of the International Conference on Technologies for Computer Electrical, Electronics & Communication (ICT-CEEL) 2023, (pp. 15-22), Khwopa.

Project

XAI User Study on LLM Integration with ML Systems **2025**

- Built a web-based Explainable AI system integrating classical ML models with an LLM-driven chatbot to explain medical recommendations, and designed a backend data pipeline to log user interactions and study outcomes for quantitative analysis.

Smart Attendance System **2023**

- Built a real-time attendance system that extracts facial features on-device and transmits embeddings to the server, reducing network load while enabling secure, proximity-based attendance verification.

Easy Read **2022**

- Developed a reading aid for visually impaired users by converting images to speech using Optical Character Recognition and Text-To-Speech, applying segmentation and noise filtering to improve text recognition accuracy by 50%.

Honors and Awards

- Recipient of the $5,000 **Graduate Research & Creative Activity (GRACA)** grant for summer research at the University of Nebraska Omaha.
- **Batch Topper Award in Bachelor's 1st, 5th and 7th semesters** for achieving the highest grade in my program.
- **First Prize Winner** in openIMIS-DRG Datathon 2022, organized by Center for Applied Research and Development (CARD), Institute of Engineering (IOE), Tribhuvan University and the Asia eHealth Information Network (AeHIN), Philippines.